MIL-G-49140(EL)
19 October 1977

MILITARY SPECIFICATION

GENERATOR, DIRECT CURRENT G-67( )/G

This specification is approved for use by Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of portable 30 watt direct current hand-cranked generator designated Generator, Direct Current G-67( )/G for charging 24 volt, vented nickel cadmium secondary batteries.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-71	Pallets, Material Handling Wood; Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	Strapping, Steel, and Seals
T-T-871	Twine, Fibrous Cotton, Wrapping
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner

Benneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, N.J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD-Form 1426) appearing at the end of this document or by letter.

FSC 6115

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PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-C-843	Cushioning Material, Celluloisic
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and Connectors)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-T-152	Treatment, Moisture and Fungus-Resistant of Communications, Electronic and Associated Electrical Equipment
MIL-P-11268	Parts, Materials, and Processes Used In Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded (Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Flexible for Packing
MIL-I-46058	Insulating Compound, Electrical (For Coating Printed Circuit Assemblies)
MIL-P-55110	Printed Wiring Boards
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging of

STANDARDS

Federal

FED-STD-356	Commercial Packaging of Supplies and Equipment

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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MIL-STD-147	Palletized Unit Loads for 40" x 48" Pallets
MIL-STD-242	Electronic Equipment Parts, Selected Standards (Forward and Contents)
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipments
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

DRAWINGS

Electronics Command

DL-SM-B-621526	Generator, Direct Current G-67( )/G

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with DL-SM-B-621526.

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4 Printed wiring boards. Printed wiring boards used in the construction of printed wiring assemblies, shall be a product which has passed the requirements of MIL-P-55110.

3.4.1 Protective coating. Printed wiring assemblies shall be protected on both sides of the printed wiring assembly with an epoxy-type conformal coating to protect the pattern against moisture, dust and other contaminants. The coating shall be Class A, Type ER in accordance with MIL-I-46058. The thickness of the coating shall be 0.0015 $\pm$0.0005 inches. For areas where protective coating cannot be applied without impairment of electrical or physical parameters, the contractor shall coordinate with the contracting officer's technical representative for procedures for masking parts before applying the coating. Hermetic sealing of the assembly in a dry atmosphere shall be acceptable in place of conformal coating.

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3.5 Marking. Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification.

3.5.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.2 Serial number. Each Generator G-67( )/G shall be serial numbered.

3.6 Preconditioning. The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.

3.6.1 Preconditioning (burn-in). Prior to delivery, and after all processing each Generator, Direct Current G-67( )/G will be burned in

3.7 Electrical requirements.

3.7.1 Rated output.

3.7.1.1 Rated output (resistive load). At a cranking speed of 60 rpm, the generator shall provide an output current of 1.0 +0.05 amperes when connected to a resistive load of 30.0 ohms. (See 4.7.1).

3.7.1.2 Rated output (battery load). With a 24 volt 5.5 ampere hour, vented nickel cadmium battery as a load, the cranking speed of the generator shall be adjusted to maintain a charging current of 1.0 +0.05 amperes to the battery. (Cranking speed range is 50 to 80 rpm.) A 10.0 ohm resistor shall be used to discharge the battery to 20.0 volts prior to the test of 4.7.1.1.

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3.7.2 Output voltage. The output voltage of the generator for rated charging current shall be from 24 to 40 volts dc over the cranking speed of 50 to 80 rpm. (See 4.7.2)

3.7.3 Overall efficiency. With the generator operating at a cranking speed of 60 rpm under rated output, the overall efficiency of the generator shall not be less than 50 percent when tested as specified in 4.7.3.

3.7.4 Overload. The generator shall be capable of withstanding repeated overloads of 2.0 amperes for one minute and cranking speeds up to 110 rpm for one minute. (See 4.7.4)

3.7.5 Reverse current. Reverse current shall not discharge the battery when connected for battery charging. (See 4.7.5)

3.7.6 Protection. All components and circuits shall be adequately rated and protected to prevent permanent damage due to external output short circuit and overloads. (See 4.7.6)

3.8 Acoustic noise. When cranked at 60 rpm under rated load the noise level shall be less than a sound level of 57 dB at 70 dB (B) weighting and 54 dB at 40 dB (A) weighting when measured as specified in 4.8.

3.9 Environmental service conditions. The equipment shall meet the following service conditions.

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3.9.1 Temperature. (See 4.9.1)

3.9.1.1 High temperature.

a. Operating. The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +125°F plus the effects of solar radiation at a rate of 360 BTU/ft/hr.

b. Storage and transporation. The equipment shall withstand exposure to ambient air temperatures as high as +160°F.

3.9.1.2 Low temperature.

a. Operating. The equipment shall be operable without degradation in specified performance at ambient temperatures as low as -50°F.

b. Storage and transportation. The equipment shall withstand exposure to ambient temperatures as low as -70°F.

3.9.2 Humidity. (See 4.9.2) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.9.3 Altitude. (See 4.9.3) The equipment shall be operable without degradation in specified performance altitudes up to 15,000 feet above sea level.

3.9.4 Vibration and shock. (See 4.9.4) The equipment shall withstand vibration and shock induced during field transport by military vehicle and rough handling. The equipment shall withstand vibrations encountered during transportation and shocks encountered during servicing.

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3.9.5 Dust. (See 4.9.5) The equipment shall withstand, in both operating and non-operating condition, exposure to fine dust particles with wind speeds of 17 knots.

3.9.6 Immersion. (See 4.9.6) The equipment shall be capable of being immersed to a covering depth of 3 feet of fresh water for a period of 2 hours without evidence of any water leakage.

3.9.7 Salt atmosphere. (See 4.9.7) The equipment shall be capable of operation in salt atmosphere as encountered in coastal areas. Prolonged periods of exposure to salt-sea atmosphere shall not result in visible evidence of corrosion on any equipment surfaces.

3.9.8 Fungus. (See 4.9.8) The equipment, in assembled and ready for delivery condition, shall provide no nutrients in material, coating or contaminent form or support fungal growth when tested in accordance with paragraph 4.9.8. In addition, the equipment shall operate satisfactorily after exposure to the test conditions of paragraph 4.9.8. Only inherently fungus resistant grades of materials per Requirement 4, MIL-STD-454, shall be used.

3.10 Electromagnetic compatibility. (See 4.10) The generator shall comply with the emission requirements of MIL-STD-461A, Notice 4 as specified below:

*RE02
*RE02.1

*Upper frequency limit of test shall be 1000 MHz. If no out of specification emissions are encountered up to 400 MHz, test may be terminated at 400 MHz.

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3.10.1 Bonds and grounds. Bonding for equipment current path return and shock hazard shall be so installed that expansion, contraction or movement incident to normal service use will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance properties from the immediate area of contact. Bonding impedance shall not exceed 2.5 milliohms. After completion of bonding installations the finish shall be restored in accordance with MIL-F-14072.

3.11 Air seal. (See 4.11) When tested as specified in 4.10 the decrease in vacuum shall not exceed 0.01 pounds per square inch, gauge (psig), per minute.

3.12 Reliability. (See 4.12) The generator shall have a specified MTBF (θ_o) of 1500 hours when operated under the following conditions:

a. Ambient temperature 95° to 113°F.

b. Operation intermittent with 5 minute cycles of continuous operation.

c. Environment - ground mobile.

3.13 Interchangeability. (See 4.13) Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be handpicked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating characteristic, etc. The equipment shall meet the requirements of MIL-STD-454 Requirement 7.

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3.14 Workmanship. (See 4.14) The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

a. MIL-STD-454 Requirement 9.

b. In MIL-P-11268:

(1) General requirements for plastic material and plastic parts.

(2) Wiring and cabling, including: Slack, protection, clearance, splicing, connections in general, grounding, shielding on wire and cable.

(3) Flux and cleaning agents for solder.

(4) Process for solder.

(5) Cleaning.

(6) Riveting.

(7) General requirements for securing of parts.

(8) Welding.

(9) Tropicalization.

(10) Mechanical operation of controls.

(11) Self-tapping screws.

c. In this Specification:

(1) Marking.

(2) Finish, protective.

(3) Printed wiring boards.

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3.14.1 Finish, protective. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as hinges, screws, etc., and necessary touch up after mounting. The final paint film on Type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.6)

3.15 Equipment safety. (See 4.14) The equipment safety features shall comply with Requirement 1 of MIL-STD-454.

3.16 Human engineering. (See 4.15) Human factors engineering and design configuration, as affecting operation, maintainability and handling shall be maintained through compliance of configuration criteria to the requirements of MIL-STD-1472.

3.17 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

(2) Quality conformance inspection of packaging. (See 4.16)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

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4.3.1 First article units. The contractor shall furnish three (3) first article units of the complete Generator G-67( )/G.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table II, and shall be performed in the order specified therein.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Parts, materials, and processes: general	3.3
Printed wiring boards and protective coating	3.4, 3.4.1
Marking	3.5

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Table II - First Article Inspection

Inspections	Reqt. Para.	Test Para.	Unit 1	Unit 2	Unit 3
1. Inspections covered by subsidiary documents		4.4	Inspections to be performed on all units.		
2. Preconditioning bounce	3.6	4.6	Inspections to be performed on all units.		
3. Group A Inspection	See Table III		Inspections to be performed on all units.		
4. Group B Inspection	See Table IV		Inspections to be performed on all units.		
5. Environmental					
Temperature tests	3.9.1	4.9.1			
Humidity	3.9.2	4.9.2			
Altitude	3.9.3	4.9.3			
Acoustic noise	3.8	4.8			
Immersion	3.9.6	4.9.6			
Dust	3.9.5	4.9.5			
Vibration	3.9.4	4.9.4.1			
Bounce	3.9.4	4.9.4.2			
Shock (drop)	3.9.4	4.9.4.3			
Fungus	3.9.8	4.9.8			
Salt atmosphere	3.9.7	4.9.7			
6. Electrical					
Protection	3.7.6	4.7.6			
Overload	3.7.4	4.7.4			
Electromagnetic compatibility	3.10	4.10			

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4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4, 4.6 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to inspections specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table III.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

Table III - Group A Inspection

Inspection	Req. Para.	Insp. Para.	AQL dphu 1/ Major	Minor
Visual and mechanical	3.14 & 3.16	4.14	4.0	6.5
Rated output	3.7.1.1	4.7.1	1.5	6.5
Output voltage	3.7.2	4.7.2	1.5	6.5
Efficiency	3.7.3	4.7.3	1.5	6.5
Air-seal	3.11	4.11	1.5	6.5
Reverse current	3.7.5	4.7.5	1.5	6.5
Protection	3.7.6	4.7.6	1.5	6.5
Overload	3.7.4	4.7.4	1.5	6.5

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4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IV and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plans, for the AQL's listed in Table IV shall be as follows:

AQL	Inspection Level
6.5	S-4

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

Table IV - Group B Inspection

Inspection	Req. Para.	Insp. Para.	AQL
Interchangeability	3.13 and 3.15	4.13	6.5
Human Engineering	3.17	4.15	6.5

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4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table V. Samples shall be selected in accordance with 4.5.3.1.

Table V - Group C Inspection

Inspection	Req. Para.	Inspe. Para.
Group 1		
Immersion	3.9.6	4.9.6
Overload	3.7.4	4.7.4
Group 2		
Shock (bench handling)	3.9.4	4.9.4.3
Temperature tests	3.9.1	4.9.1
Group 3		
Altitude	3.9.3	4.9.3
Dust	3.9.5	4.9.5
Acoustic noise	3.8	4.8
Salt atmosphere	3.9.7	4.9.7
Humidity	3.9.2	4.9.2
Vibration	3.9.4	4.9.4.1
Bounce	3.9.4	4.9.4.2
Electromagnetic compatibility	3.10	4.10
Group 4		
Shock (drop)	3.9.4	4.9.4.3
Fungus	3.9.8	4.9.8

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4.5.3.1 Group C sampling.

4.5.3.1.1 Sampling for inspection (Group 1). One sample shall be selected at random, from every 50 equipments or fraction thereof produced and subjected to each inspection specified. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.2 Sampling for inspection (Group 2). Two samples shall be selected at random each month and subjected to each inspection specified. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.3 Sampling for inspection (Group 3). One sample shall be selected at random, from each 200 equipments produced or fraction thereof, and subjected to each inspection specified. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3) An additional unit shall be selected for EMI/EMC tests only.

4.5.3.1.4 Sampling for inspection (Group 4). This inspection shall be required only when First Article samples are not specified by contract or purchase order. When required, one sample shall be selected at random, at the start of the contract, from the first quality conformance inspection production lot (See 4.5.3), and subjected to each inspection specified.

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4.5.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.3.3 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.4 Group D inspection. This inspection shall consist of the tests specified in Table VI and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

Table VI - Group D Inspection

Inspection	Req. Para.	Insp. Para.
Reliability	3.12	4.12

4.5.4.1 Sampling for Group D Inspection.

4.5.4.1.1 Sampling for First Article Reliability Qualification. For the purpose of First Article Qualification, 10 units shall be tested.

4.5.4.1.2 Sampling for Quality Conformance (Group D) Inspection. For the purposes of Quality Conformance Inspection 10 units shall be selected at random for each lot. Unless otherwise specified for reliability testing purposes, a lot is defined as the greater of one months production or 60 units.

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4.5.4.2 Group D failures. In the case of Group D failure occurrence, procedures and requirements specified elsewhere in the ocntract shall be followed.

4.5.4.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and passed Group A inspection.

4.6 Bounce preconditioning. The unit, shall be placed in its normal operation position on the table of the Type 1000-SC Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.6.1 Preconditioning. Prior to submission to the specified quality conformance inspections, each unit shall be operated for a minimum of 48 hours under the conditions specified for reliability testing. The preconditioning period (burn-in) shall be concluded by a failure free period, the duration of which shall be the greater of 24 hours or one-third of the total burn-in time.

4.7 Electrical tests. The appropriate electrical tests shall be performed with the generator driven by a motor drive and suitable gear reduction to drive the generator. The motor drive shall be capable of a maximum cranking speed of 110 RPM and be capable of maintaining the specified speed within -0 +5 RPM.

4.7.1 Rated output. (Resistive load). With the generator connected to a resistance load of 30.0 ohms and cranked at a speed of 60 RPM the output current shall be measured with an external ammeter and shall meet the output requirements of 3.7.1.1

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4.7.1.1 Rated output (battery load). With the generator connected to a 24 volt, (discharged to 20.0 volts) 5.5 ampere hour, vented nickel cadmium battery, such as Type BB-651, as a load, the charging current shall be measured with an external ammeter and shall meet the requirements of 3.7.1.2. During this test the cranking speed shall be adjusted between 50 to 80 RPM to maintain an output charging current of 1.0 ±0.05 amperes. The charge cycle shall continue for a period of eight (8) hours or until a battery voltage of 30.0 volts is reached.

4.7.2 Output voltage. The generator will be cranked at any rate between 50 RPM and 80 RPM, adjusting the load resistor at each cranking rate to deliver a load current of 1 ±0.05 Amp. The output voltage shall be measured with an external voltmeter and shall be 24 Vdc $\geq E_{out} \geq$ 40 Vdc as required by paragraph 3.7.2.

4.7.3 Efficiency. The efficiency of the generator shall be measured as follows: The generator shall be mounted to swing freely on its crankshaft. A beam with an effective leverarm exactly 1 foot long measured to the center of the crankshaft, and at right angles to the side of the generator, shall be attached to that side of the generator where the torque will tend to move the beam upward. After balancing the generator so that the beam is in a horizontal position, the generator shall be under rated load current of 1.0 ±0.05 amperes and at a speed of 60 rpm using a motor and a suitable gear reduction to drive the generator. The vertical restraining force at the end of the beam required to maintain the beam in a horizontal position shall be measured. Then:

Percent efficiency of the generator-

$$\frac{7.04 \times W}{FN} \qquad 100$$

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Where W= Watts output of the generator

F= Pounds restraining force required to prevent rotation of the beam

N= Number of revolutions per minute of the crankshaft

7.04 = Constant

A strain gage type torque indicator may be substituted for the leverarm and force measuring device.

4.7.4 Overload. The generator shall be driven by a motor drive at a cranking speed of 110 rpm. A resistance load adjusted to obtain an output current of 2.0 ±.0.05 amperes shall be connected across the output of the generator for a period of one (1) minute. The load shall then be removed and the cycle repeated ten(10) times allowing 5.0 minutes between tests. The motor drive shall be capable of maintaining the cranking speed within -0 +5.0 rpm during this test. After completion of the test the generator shall show no evidence of mechanical or electrical failure and shall meet the requirements of 3.7.1.1, 3.7.2 and 3.7.4.

4.7.5 Reverse current. A dc power supply with an adjustable output voltage from 0 to 50 Vdc shall be connected to the output terminals of the generator in correct polarity (with the generator not being cranked, and the meter switch in the current position). The voltage of the power supply shall be gradually increased to 50 volts dc. The polarity of the power supply shall then be reversed and the test repeated. The leakage current to the generator shall not exceed 0.5 ma. for the correct polarity condition and 150 ma. for reverse polarity. After the test the generator shall show no damage and shall meet the output requirements of 3.7.1.1, 3.7.2 and 3.7.3.

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4.7.6 Protection. With the generator being cranked at a speed of 60 rpm a short circuit shall be momentarily applied across the output terminals. The output short circuit shall be applied a minimum of 5 times allowing one (1) minute between cycles. During the short circuit period the cranking speed of the generator shall not be less than 55 rpm. Adter the test, the generator shall show no damage and shall meet the requirements of 3.7.1.1, 3.7.2 and 3.7.3.

4.8 Acoustic noise. The acoustic noise due to operation of the generator, shall be measured with a sound level meter (type 1551-B as manufactured by General Radio Co., Cambridge, Mass., or equal) with B-weighting (70 db) and A-weighting (40db) in an open area. These measurements shall be made at a distance of three (3) feet from the top front edge of the generator to the center of the sound level meter microphone and with the generator mounted for operation on the ground and to a fixed object. With the generator operating at a cranking speed of 60 rpm under rated load, the acoustic noise level of the generator shall not exceed the values specified in 3.8.

4.9 Environmental service conditions. The following tests shall be performed on the equipment. When a test requires measurements to be made before, during or after the test, these measurements shall be accomplished as specified in 4.7.1.1, and 4.7.2.

4.9.1 Temperature.

4.9.1.1 High temperature. The equipment shall be subjected to the test of Method 501.1, Procedure II of MIL-STD-810. For Step 7, the chamber temperature shall be adjusted to 71°C (160°F) for test operation. (This includes effects of solar radiation).

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4.9.1.2 Low temperature. The equipment shall be subjected to the test of MIL-STD-810, Method 502.1, Procedure I. The temperature in Step 2 shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (Step 4) shall be -45.6°C (-50°F).

4.9.2 Humidity. The equipment shall be subjected to the test of Method 507.1, Procedure III of MIL-STD-810. Measurements shall be made during the last 5 hours of the last cycle of Step 6, and during the last 5 hours of the steady-state exposure in Step 8.

4.9.3 Altitude. The equipment shall be subjected to the test of MIL-STD-810, Method 500.1, Procedure I. The test shall include a 40,000 foot altitude non-operational exposure for not less than 1 hour previous to the 15,000 foot altitude operational portion of Step 2. Altitude rate charges shall not exceed 2,000 feet per minute.

4.9.4 Vibration and shock.

4.9.4.1 Test for vibration. The equipment in its combat transportation mode shall be subjected to the test of MIL-STD-810, Method 514.2, Procedure X. The vibration level shall be in accordance with Table 514.2-VII, Curve AW, (Figure 514.2-7).

4.9.4.2 Bounce. The equipment shall be subjected to the test of MIL-STD-810, Method 514.2, Procedure XI, Part 2. Prior to and following this test the equipment shall be operated in accordance with MIL-STD-810, General Requirements paragraph 3.2. The equipment shall not have any degradation in performance. The equipment shall not suffer any mechanical damage.

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4.9.4.3 Shock. The equipment in its transportation mode shall be subjected to the test of MIL-STD-810, Method 516.2, Procedure II and V.

4.9.5 Dust. The equipment shall be subjected to the test of Method 510.1, of MIL-STD-810. The "Generator" shall operate in Step 4 immediately after turning off the chamber controls. Accumulated dust may be removed from the test item prior to operation. Following return to standard ambient conditions, the test item shall again be operated. The orientation of the Generator shall be changed every 1-1/2 hours in Step 1 so that the four sides of the equipment face directly into the dust stream. Step 2 conditions shall be held for 16 hours.

4.9.6 Immersion. The equipment as prepared for field transport (in it's transit case, if any), shall be subjected to the test of Method 512.1, Procedure I, of MIL-STD-810. This test shall be performed prior to and following vibration and shock testing of the same test item.

4.9.7 Salt atmosphere. The equipment shall be subjected to the test of Method 509.1, Procedure I, of MIL-STD-810. No electrical operation shall be required, but moving parts not sealed against the atmosphere shall be operated through their full range of movement or adjustment. If the test item is unable to perform its intended function, movement, or adjustment, it shall have failed the salt atmosphere test. In addition if the test item shows obviously flaked, pitted, blistered, delaminated, or otherwise loosened or corroded base material, surface finish or treatment of internal or external surfaces it shall have failed the test.

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4.9.8 Fungus. The equipment shall be subjected to Test Method 508.1, Procedure I of MIL-STD-810. Operation of the equipment shall not be required. At the conclusion of the exposure period, examination shall reveal no evidence of viable fungus or corrosion on any surface.

4.10 Electromagnetic compatibility. The equipment shall be tested for compliance to the requirements of 3.10 of this specification. Test set ups and procedures employed shall conform to the measurement techniques of MIL-STD-462, Notice 3.

4.10.1 Bonds and grounds. Prior to the performance of EMI-EMC tests the bonding impedance requirements of 3.10.1 shall be verified. Bonding impedance of the following items shall be measured with a Shallcross Meter Model 670A, or equal:

a. Electrical bonds and grounds.
b. All surfaces requiring EMI gaskets.
c. All switches and meter cases to commom ground.

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4.11 Air-seal test. The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment inclosure. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed .01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of .01 pound can be determined readily. (See 3.11).

4.12 Reliability testing.

4.12.1 First article reliability test. The First Article reliability inspection testing shall be in accordance with Test Plan IV of MIL-STD-781.

4.12.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan IVA of MIL-STD-781.

4.12.3 Reliability test conditions. The reliability test shall demonstrate a specified MTBF (θo) as required in paragraph 3.12. The test shall be run according to the conditions of test Level B of MIL-STD-781 except equipment cycling shall be limited to 5 minutes on, 1 minute off. Each unit shall be connected to a resistance load of 30.0 ohms and a cranking speed of 60 rpm. A failure for reliability purposes shall be defined as the inability of any equipment to meet the requirements of this specification. Each unit shall be continuously monitored to detect any failures. The total test time for each equipment shall be in accordance with paragraph 4.2.2.4 of MIL-STD-781. Paragraph 5.5 thru 5.9 of MIL-STD-781 shall apply.

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4.13 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.13. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable subassembies and replaceable parts, when such dimensions affect mating of parts.

b. Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

c. Location of fasterners on separate parts or assemblies which must mate together such as cases, covers, and mountings.

4.14 Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252. An inspection shall be performed to verify compliance with those parts of paragraph 3.15 which can be determined visually.

4.15 Inspection for human engineering. Human engineering verification and maintenance of design configuration to the requirements of paragraph 3.16 shall be verified by visual and mechanical inspections.

4.16 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.16.1 Rough handling test (packaging). When the rough handling test in accordance with MIL-P-116 is required by contract (see 6.2.f) the following functional test shall be conducted to determine freedom from operational malfunction caused by rough handling:

Inspection	Measurement	Cranking Speed
3.7.1	4.7.1	60 RPM

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A, B or Commercial, as specified (see 6.2b).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Generator, Direct Current G-67( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Generator, Direct Current G-67( )/G shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be individually unit packed Method IC-1.

5.1.1.4.2 Cable assembly, battery charging. Each cable assembly shall be individually unit packed in accordance with MIL-C-55442.

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5.1.1.4.3 Generator, Direct Current G-67( )/G. Each generator shall be individually unit packed Method III as follows: The strap shall be wound and secured with twine conforming to T-T-871, Type II. Place the generator and secured strap together with the cable assembly, unit packed as specified in 5.1.1.4.2, into the generator carrying bag. Fill voids with cushioning material conforming to PPP-C-843, Type II, and secure flap with fasteners. The carrying bag with contents inclosed shall be cushioned on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, Type CF, class weather-resistant, variety SW, Grade W5c, designed to protect the unit from shock and vibration hazards during handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. The cushioned item shall be inclosed within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservation application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Prevervation shall be in accordance with FED-STD-356.

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5.2 Packing. Packing shall be Level A, B or Commercial, as specified (see 6.2b). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Generator, Direct Current G-67( )/G, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, Style 4, Class 2; or PPP-B-585, Style 2 or 3, Class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, Type I, Finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load Type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, Type IV, Group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, Type I, Finish A, or shrink film.

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5.2.2 Level B. A quantity of Generator, Direct Current G-67( )/G, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, Class 2, Style E, or PPP-B-636, Type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or non-metallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, Group I or II woods. The load shall be "bonded" to the pallet by Strapping conforming to QQ-S-781, Type I, Finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Generator, Direct Current G-67( )/G, unit packed as specified in 5.1, shall be placed on a pallet, load Type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, Group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, Type I, Finish A, or shrink film.

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5.2.3 Commercial packing. Packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

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6. NOTES

6.1 Intended use. Generator G-67( )/G is intended for charging vented nickel-cadmium secondary batteries used with portable communication equipments such as Radio Set AN/PRC-70.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level A, Level B or Commercial preservation and packing (see Section 5).

c. First article inspection:

(1) Three (3) sample units of each item cited in Section 1 are generally required so that lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on the second sample unit. (See 3.1).

(2) When pre-production pack is required, one acceptable model of Generator, Direct Current G-67( )/G shall be packed in a manner that will pass the first article inspection prescribed in MIL-P-116 (see 4.16). Illustrations of the packing procedure, together with Bills of Material, shall be prepared and furnished in accordance with the provisions of Signal Corps Drawing No. SC-A-362100.

d. Marking and shipping of samples.

e. Place of final inspection.

f. When rough handling test is required. (See 4.16.1).

g. Submission of the statement of treatment covering tropicalization of material as soon as possible after award of contract. This statement should be submitted to the contracting officer.

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6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: G-67W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 Group C and D inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2 and 4.5.4.2).

6.6 Color. The color chip furnished by the procuring agency (see 3.14.1) will match color chip No. X-24087 and may be obtained upon request to Command Officer, U.S. Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA. 19120.

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6.7 Location of air-seal test. It is desirable that the air-seal test (4.11) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.8 Definitions.

6.8.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.8.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.8.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

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6.8.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.8.5 Tublar germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.9 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-G-49140(EL) GENERATOR, DIRECT CURRENT G-67( )/G

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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